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GeNIS: A modular dataset for network intrusion detection and classification

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ABSTRACT

The development of artificial intelligence solutions for cyber-attack detection and classification require high-quality and representative data. However, there is a scarcity of labelled datasets focused on the cyberattacks that target vulnerable small and medium-sized enterprises. To allow organizations to improve their intrusion detection systems according to their types of users, their active services, and the network protocols they use, it is necessary to provide reliable captures of different types of benign and malicious traffic. The GECAD Network Intrusion Scenarios (GeNIS) dataset contains multiple sequential attack scenarios and different types of realistic normal network activity, recorded during advanced network simulations on the Airbus CyberRange platform. The raw network packets were analyzed to generate labelled network flows, with the computation of statistical features to represent the traffic patterns of local and remote attackers, normal users and administrators, and background traffic of an enterprise computer network. GeNIS follows a modular design, providing raw packet capture next generation (PCAPNG) files with over 37 million packets of each intermediate attack step to enable an in-depth analysis with different flow exporters, feature extraction, and feature selection tools, as well as filtered CSV files with over 2.8 million flows created with 5, 10, 30, and 60 s flow intervals. The flows were pre-processed to provide a reliable benchmark dataset with the

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Table 3
Admin profile activity schedule.

Hours	Protocol
9:00	–
10:00	SSH, FTP, HTTP, SMB
11:00	SSH, DNS
12:00	–
13:00	FTP
14:00	HTTP
15:00	SSH, SMB
16:00	SSH

dataset also includes machine background traffic, which was passively captured. This ensures a more comprehensive representation of the device's overall network behavior.

Profile 2 involves various administrative tasks performed by system administrators. These tasks include uploading files to FTP servers, accessing remote machines via SSH, testing DNS functionality, manually interacting with SMB administrative shares to add or retrieve configuration files, and testing the functionality of web servers. Table 3 provides a summary of the traffic generated by this profile, as configured in the BUP.

The traffic generated by the two profiles was captured separately to provide a modular approach for traffic analysis. In addition to the traffic generated by the BUP profiles, there is also background traffic generated by the processes inherent in the operation of the servers and clients. This background traffic includes ARP packets, NBNS packets related to Active Directory functionality, and TCP and DNS traffic. This background traffic was captured separately from the BUP-generated traffic to allow for an analysis of the network traffic prior to applying the BUP.

4.3. Malicious traffic

The malicious network traffic generated is divided into eight different scenarios, each representing a specific attack strategy to simulate realistic cyber threats. The attacker, who is located on the Kali-User and Kali-Remote devices within the user and remote LAN, executes the attacks in a modular, step-by-step manner. Each step builds on the previous one, reflecting the progressive nature of real-world cyberattacks. This structured methodology captures the sequential dependencies of multi-stage attacks such as reconnaissance, exploitation, and disruption, enabling detailed analysis of individual attack steps or the entire attack flow. A pause of approximately five minutes is built in between each step, allowing the attacker to analyze the results of the previous step before continuing.

4.4. Scenario 1

This scenario illustrates an attempt to disrupt business operations by slowing or denying access to a business-critical service. Here, the attacker targets the web server, which is a vital asset to the organization's operations. The steps in this scenario are shown in Table 4.

The scenario starts with two attackers: one on the Remote User's LAN and one on the User's LAN. Both exploit the DNS configuration of the environment, which resolves the domain name of the web server as "web.sigen.net". This allows the attackers to identify the corresponding IP address of the server. Next, the attackers use NMAP [10] to scan the target and find the port on which the web service is running. With this information they launch a Hulk [11] HTTP flood. The remote attacker initiates their steps as the Kali-User is in the middle of the attack, flooding the server with a massive amount of traffic. The overwhelming influx of requests causes the server

Table 4
Scenario 1 attack sequence.

SC1 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
DoS Hulk

Table 5
Scenario 2 attack sequence.

SC2 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
DoS Slowloris

to experience severe performance degradation, potentially rendering the service completely inaccessible.

4.5. Scenario 2

Similar to Scenario 1, this scenario illustrates an attempt to disrupt business operations by either slowing or blocking access to a critical service, in this case the web server, for a long period of time. The sequence of events in this scenario is shown in Table 5.

As in Scenario 1, two attackers are involved: one on the Remote User's LAN and one on the User's LAN. Both exploit the environment's DNS configuration, which resolves the web server's domain name to "web.sigen.net," allowing the attackers to obtain the server's IP address. The attackers then use NMAP to scan the target and identify the port on which the web service is running. Once this information is obtained, they launch a Slowloris attack using Metasploit [12], which keeps connections open for long periods of time to exhaust server resources. The remote attacker initiates his part of the attack in the middle of the Kali-User actions, overwhelming the server with a large volume of traffic. This surge of traffic causes severe performance degradation, potentially rendering the service completely inaccessible.

4.6. Scenario 3

This scenario illustrates an attempt to disrupt business operations by slowing or blocking access to a critical service. The attacker focuses on the web server, which is a key asset for the organization to function. The sequence of actions in this scenario is shown in Table 6.

Table 6
Scenario 3 attack sequence.

SC3 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
DoS UDP

Table 7
Scenario 4 attack sequence.

SC4 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
DoS ICMP

The attacker, located on the User's LAN, exploits the environment's DNS configuration, which resolves the Web server's domain name to "web.sigen.net". This allows the attacker to determine the IP address of the server. The attacker then uses NMAP to scan the target and identify the port on which the web service is running. With this information, they initiate a UDP flood attack using Hping3 [13], which compromises the server with a high volume of traffic.

4.7. Scenario 4

This scenario depicts an attempt to disrupt the normal operations of an organization by degrading or blocking access to a critical service. As previous scenarios, the attacker targets the web server, which is an essential asset for the organization's day-to-day activities. The series of steps involved in the attack is shown in Table 7.

The attacker, located on the User's LAN, exploits the DNS configuration that resolves the Web server's domain name to "web.sigen.net", allowing the attacker to find the IP address of the server. With this information the attacker uses NMAP to scan the target and identify the port on which the web service is running. Once this is discovered, the attacker launches an ICMP flood attack using Hping3, flooding the server with a massive amount of traffic.

4.8. Scenario 5

This scenario illustrates an attempt to disrupt the operation of an organization by limiting or preventing access to a critical service, the web server. The sequence of actions in this attack is detailed in Table 8.

Following a pattern similar to the previous scenarios, the attacker, positioned on the User's LAN, exploits the DNS configuration to identify the IP address of the web server. Using this information, the attacker performs an NMAP scan to locate the port on which the web service is running. Once the port is identified, the attacker initiates a Push&Ack flood attack using Hping3, flooding the server with an overwhelming amount of traffic.

Table 8
Scenario 5 attack sequence.

SC5 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
DoS Push&Ack

Table 9
Scenario 6 attack sequence.

SC6 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
Bruteforce SMB

Table 10
Scenario 7 attack sequence.

SC7 sequence
DNS exploitation
5 min pause
NMAP mapping
5 min pause
Bruteforce SSH

4.9. Scenario 6

This scenario outlines an attack that uses a bruteforce SMB attack to test credentials in order to gain unauthorized access to network resources. By systematically trying a large number of potential login combinations, the attackers attempt to identify valid credentials that will allow them to escalate privileges, move laterally within the network, and ultimately compromise critical systems or sensitive data. The sequence of actions in this scenario is shown in Table 9.

Similar to previous scenarios, the attacker begins by identifying the IP address of the DNS server. During an NMAP scan, they discover that the DNS server has an active SMB service running. Taking advantage of this, the attacker then deploys Hydra [14], a password cracking tool, using a dictionary of the 10,000 most common passwords [15]. By bruteforcing AD credentials, the attacker can gain unauthorized access to the machine, which opens the door for further malicious activity within the network.

4.10. Scenario 7

This scenario depicts an attack designed to gain unauthorized access to a computer and its files. Such access could expose sensitive data or serve as a first step to further exploitation, such as malware injection, ransomware, or insider information gathering. The detailed steps are shown in Table 10.

Similar to the previous scenarios, the attacker first determines the IP address of the machine hosting the web site. During an NMAP scan, the attacker discovers an active SSH service on the target. Using this discovery, they deploy Hydra to try and crack the password, using a dictionary of the 10.000 most commonly used passwords. By bruteforcing the SSH credentials, the attacker can gain unauthorized access to the machine, setting the stage for further malicious activity.

4.11. Scenario 8

Similar to scenario 7, this scenario involves attacking a server to gain unauthorized access to sensitive files. Like so, the breach could lead to subsequent malicious actions such as injection of malware or deploying ransomware. In addition, the attacker could gain critical insider knowledge, such as database credentials or system configurations. The steps are shown in Table 11.

Table 11
Scenario 8 attack sequence.

SC8 sequence
DNS exploitation
5 min pause
NMAP reconnaissance + mapping
5 min pause
Bruteforce SSH + FTP

Following a pattern similar to the previous scenarios, the attacker first identifies the IP address of the Web server. Knowing that this server is part of the DMZ LAN, they use NMAP to scan the LAN for active machines and services. Among the devices identified, an FTP server that hosts both FTP and SSH services stands out. The attacker then performs bruteforce attacks on both services using a password dictionary containing the top 10.000 commonly used passwords and Hydra. These attacks could provide access to sensitive data or a leverage point for further compromises.

4.12. Feature extraction

Concerning the raw network packets generated from the previously described traffic generation steps, HERA [16], a tool designed to streamline the creation of flow-based datasets, was selected to aggregate these packets into flows and extract the features to construct the final dataset. An advantage this tool provides is the possibility of generating flows in different intervals, specifically the frequency in seconds at which a flow will be generated as long as there is still on-going activity in the flow during the specified time. This is a useful functionality when considering smaller intervals, such as the minimum of 1 s, will provide more updates on the flow while a higher number will generate fewer flows and allow for more summarization.

HERA was used to generate different versions of the dataset with flow intervals of 5, 10, 30 and 60 s, and then saved them in a CSV file format, where each line represents a network flow with multiple extracted features that summarize network statistics. Then, the same tool was used to label the dataset, categorizing each flow as either benign or belonging to a specific attack type as identified in the previous sections. Regarding the labeling process, it relied on a ground truth file that was carefully constructed to contain identifiable information about each

Table 12
Dataset statistics.

Flow Volume Metrics		5 s	10 s	30 s	60 s
Total number of packets		37.681.001			
Total number of bytes		2.737.930.223			
Flows	dos-udp	786.432	393.216	131.072	65.536
per label	dos-icmp	786.432	393.216	131.072	65.536
	dos-pushack	785.472	392.736	130.942	65.806
	dos-slowloris	255.311	168.116	85.212	51.729
	dos-hulk	93.938	67.821	51.327	47.033
	recon-nmap	27.713	27.713	27.713	27.713
	benign-background	20.649	17.180	12.810	10.286
	benign-users	20.355	18.048	15.641	12.851
	bruteforce-ssh	12.388	8.677	4.696	4.688
	bruteforce-smb	10.002	10.001	10.001	10.001
	benign-admin	4.112	4.096	4.083	4.013
	bruteforce-ftp	3.344	3.344	3.344	3.344
	recon-dns	20	20	20	20
Total number of flows		2.806.168	1.504.184	607.933	368.556

attack instance that would allow us to properly classify the traffic. This included timestamps marking the beginning and end of the attack, IP addresses, port numbers, and the relevant protocols. To ensure the accuracy of the ground truth and to ensure that a reliable dataset was produced, a multi-step validation process was conducted. First, the attack scenarios were reviewed to confirm the expected malicious activity. Then, a thorough analysis of the raw traffic within the capture files was performed, cross-referencing known attack behavioral patterns to identify malicious traffic. Table 12 presents a summarized view of the traffic statistics found in the dataset.

Limitations

Despite the careful design of this dataset, it is important to acknowledge its limitations. Since the performed cyberattacks were restricted to specific machines, expanding the scenarios to include more types of attacks across different network protocols and more LANs could improve data diversity. Additionally, even though the dataset includes benign network activity, it does not fully capture the variability observed in complex computer networks. Combining this dataset with real-world benign traffic generated by different user behaviours across multiple applications and device types could improve the generalization of machine learning and deep learning models.

Ethics Statement

The authors confirm that they have read and followed the ethical requirements for publication in Data in Brief. This work does not involve human subjects, animal experiments, or any data collected from social media platforms.

CRediT Author Statement

Miguel Silva: Methodology, Data curation, Validation, Investigation, Writing. **Daniela Pinto**: Methodology, Data curation, Validation, Investigation, Writing. **João Vitorino**: Conceptualization, Methodology, Validation, Investigation, Writing. **José Gonçalves**: Methodology, Validation, Investigation, Writing. **Eva Maia**: Conceptualization, Validation, Investigation, Supervision. **Isabel Praça**: Conceptualization, Resources, Project administration, Funding acquisition.

Data Availability

GeNIS: GECAD Network Intrusion Scenarios (Original data) (Zenodo).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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